Programming Project #3
Integer Sets

1 Overview

Write the CADT module `intSet` to implement an ordered set of integers according to the specification given below.

File `intSet.h` (downloadable off the class web pages):

```
#include <stdbool.h>


#ifndef INTSET_H
#define INTSET_H

typedef struct intsetType *intSet;

intSet createSet();      // returns an intSet created at runtime using malloc
void destroySet(intSet); // frees up the memory associated with its argument
void clear(intSet);      // clears set to empty (does not free up memory)
int card(const intSet);  // returns the cardinality of the set
bool equals(const intSet,const intSet); // returns true if 2 sets are equal, false
                                        // otherwise; two sets are equal if they contain
                                        // all of the same values
bool contains(const intSet,int); // retuns true if argument is in set, false otherwise
int largest(const intSet);   // returns the largest item in the set; prints error message
                             // and exits if set is empty
int smallest(const intSet);  // returns the smallest item in the set; prints error message
                             // and exits if set is empty
void add(intSet s,int item);     // adds item to s or does nothing if it is in set
void remove_(intSet s,int item); // removes item from s or does nothing if it is not in set

intSet union_(const intSet, const intSet);     // set union
intSet intersect(const intSet, const intSet);  // set intersection
intSet diff(const intSet s1, const intSet s2); // set difference, i.e. s1 - s2

bool isEmpty(const intSet); // returns true if the set is empty, false otherwise

int *toArray(const intSet);   // returns an array representation of the set in
                              // ascending order
char *toString(const intSet); // returns a string representation of the set in
                              // ascending order;  e.g. an empty set returns "{}", and
                              // a non-empty set containing the values 9,-52,17 would
                              // return the string "{-52,9,17}"

#endif
```

You may not modify file `intSet.h` in any way. Note that you must define `intsetType` in your C implementation file as a structure. Two of the routines (`remove_`, `union_`) contain an underscore

(_) in the name because they conflict with a keyword in C (`union`) and a function in the standard libraries (`remove`).

Notes on the project:

1. You must store your set in an array that can expand as needed to use all of available memory.
2. A parameter of type `const` means you should not modify the argument in any way.
3. You may not use any external C set libraries, set routines, or container libraries for the project. You also may not use any external sort routines, including the ones in the C standard library.
4. Your error messages must contain the string "error" (case is irrelevant).
5. Your module should only export or make public the routines defined in the specification file.
6. Set difference: $A - B \equiv \{x : x \in A \text{ and } x \notin B\}$

2 Suggested Program Development

Start by defining your data structure(s) and `createSet`. Note that how you define your data structure(s) will strongly influence how you will need to write some of your functions. You should probably implement at least a first version of `add` next so you can begin to build a set, followed closely by either `toArray` or `toString` so you have the capability to actually inspect any `intSet` you create. **Test and implement any new function you write completely before moving on to additional routines.** Always remember that any routine defined in your module can be used in any of the other routines.

3 Deliverables

You should only turn in your `intSet` implementation file! I will test your code with my own driver test program(s). You may work in pairs on this project; be sure to submit only one version of your code plus have have all names in the comments at the beginning. Submit your programs via Canvas in the usual manner. Note you need not turn in an executable file or your driver program!